



Lesson 6.7 — Solving Exponential Equations

Big idea: If you can rewrite both sides with the same base, equate the exponents. If you can't, take a log of both sides and solve linearly.

Quick review

Same base method: $a^{f(x)} = a^{g(x)} \Rightarrow f(x) = g(x)$.

Different bases method: take $\log$ (or $\ln$) of both sides, then use the power rule to bring the variable down.

Watch units / domain: exponential output is always positive, so the side after isolating must be positive.

When you have a coefficient: isolate the exponential first, then take the log.

Worked example

Worked example. Solve $5 \cdot 2^x = 80$.

Divide by 5: $2^x = 16 \Rightarrow x = 4$. (Or take $\log_2$: $x = \log_2(16) = 4$.)

Warm-ups

1. Solve $3^x = 81$.
2. Solve $2^x = 1/8$.
3. Solve $5^x = 1$.

Practice

1. Solve $4^{x+1} = 64$.
2. Solve $7^x = 50$ (decimal answer to 4 places).
3. Solve $10^{2x} = 1000$.

Challenge

1. Solve $3 \cdot 2^x = 7 \cdot 5^x$.
2. An investment doubles every 9 years. How long does it take to triple? (Round to nearest year.)



Answer key — Lesson 6.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x = 4$ ($3^4 = 81$).
2. $x = -3$ ($2^{-3} = 1/8$).
3. $x = 0$.

Practice

1. $4^{x+1} = 4^3 \Rightarrow x + 1 = 3 \Rightarrow x = 2$.
2. $x = \log 50 / \log 7 = \ln 50 / \ln 7 \approx 3.9120/1.9459 \approx 2.0103$.
3. $10^{2x} = 10^3 \Rightarrow x = 3/2$.

Challenge

1. Take $\ln$: $\ln 3 + x \ln 2 = \ln 7 + x \ln 5 \Rightarrow x(\ln 2 - \ln 5) = \ln 7 - \ln 3 \Rightarrow x = \ln(7/3)/\ln(2/5) = (0.8473)/(-0.9163) \approx -0.9247$.
2. $A(t) = A_0 \cdot 2^{t/9}$. Triple: $3 = 2^{t/9} \Rightarrow t/9 = \log_2 3 \approx 1.585 \Rightarrow t \approx 14.27$ years ≈ 14 years.